

No. of Functional Features Included in the Solution

Date	1 October 2026
Team ID	SWTID-2026-3138
Project Name	ComicCraft - AI Comic Story Creator using Gemini Models
Maximum Marks	5 Marks

Functional Features Overview:

This document lists all the functional features included in the project solution. Each feature is described with its purpose, implementation status, and contribution to the overall solution.

S.No	Feature Name	Feature Description	Module / Component	Status (Done / In Progress / Pending)	Marks Contribution
1	User Input Form	Form to enter story prompt, main character name, setting, story tone and art style; submits via POST to /generate.	index.html (Jinja2 + CSS)	Done	0.5
2	AI Panel Outline Generation	Gemini 1.5 Flash creates a structured 5-panel outline (panel no., title, scene description, image prompt) as validated JSON.	gemini_flash.py - generate_outline()	Done	0.75
3	Story Narration & Dialogue	Gemini 1.5 Pro expands the outline into engaging narration, captions and character dialogue for every panel.	gemini_pro.py - generate_story()	Done	0.75
4	Comic Image Generation	Stable Diffusion v1.5 (Diffusers) draws one comic-style image per panel; filename sanitized, saved to static/panels.	image_generator.py - generate_image()	Done	0.75
5	Comic Layout Builder	Matches each image with its panel story and builds a layout list (panel no., title, image path, text, scene description).	layout_builder.py - build_comic_layout()	Done	0.5
6	Comic Preview Page	Shows the comic panel by panel: title, illustration, scene description (italics), caption and narration; has the PDF download button.	comic_preview.html + /generate route	Done	0.5
7	PDF Export & Success Page	Compiles images and narration into a multi-page timestamped PDF using FPDF (static/exports); redirects to the export success page.	exporters.py - save_pdf(); export_success.html	Done	0.75
8	JSON API & Image Test Endpoints	/generate-comic/json returns layout data and PDF path for API clients (Pydantic PromptRequest); /test-image tests image generation alone.	routes.py (FastAPI)	Done	0.5

Total marks contribution: 5.0 / 5

Feature Summary:

Metric	Count / Value
Total Features Planned	8
Total Features Implemented	8
Core / Must-Have Features	7 (Features 1 to 7 - the complete prompt-to-PDF workflow)
Additional / Nice-to-Have Features	1 (JSON API and image test endpoints)
Features Tested & Verified	8 (verified through local deployment testing, Milestone 5.2)

Feature Category Breakdown:

S.No	Category	Features in Category	Example Features
1	User Interface (UI)	2	Input form (index.html); comic preview and export success pages (Jinja2 templates)
2	Backend / Logic	2	Layout builder; PDF export logic; FastAPI routing and input validation
3	Database / Storage	0 (file storage only)	No database used; panel images saved in static/panels and PDFs in static/exports
4	API / Integration	4	Gemini 1.5 Flash, Gemini 1.5 Pro, Stable Diffusion v1.5, JSON API /generate-comic/json
5	Security / Authentication	0 (supporting only)	API keys in .env (GEMINI_API_KEY, HF_API_KEY); form and Pydantic input validation. No user login in this version